

Seth DeWhitt

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EDUCATION

Purdue University, *West Lafayette, IN*

August 2023 – May 2027

B.S. in Computer Science, Software Engineering concentration

GPA: 3.6/4.0

Relevant Coursework: Data Structures, Analysis of Algorithms, Software Engineering, Systems Programming, Computer Architecture, C Programming, Software Testing, Object-Oriented Programming, Linear Algebra

Activities: Purdue Jazz Education Network (President), Purdue Jazz Band (Lead Tenor Saxophone)

Awards: Elmhurst University Jazz Festival Outstanding Soloist (4x), Dean's List (3x)

EXPERIENCE

Research Lead & Teaching Assistant

August 2025 – Present

The Data Mine, Purdue University | Agile, Python, Typescript, Surreal, Firebase

West Lafayette, IN

- Directing 11 student researchers in collaboration with the Practical AI Podcast (**1.5M listens**)
- Implementing RAG chatbot trained on **349+ episodes** with citation-backed responses
- Designed scalable podcast generation platform integrating LLM-produced transcripts and voice clone models

Software Engineering Intern

June 2025 – August 2025

Chewy | Python, React, Vite, FastAPI, Docker, AWS, Terraform, Jenkins, PostgreSQL

Bellevue, WA

- Deployed scalable & containerized development environments with custom tooling for in-house model execution on AWS ECS/Fargate, **reducing setup overhead by 80%** and supporting **15+ engineers**
- Automated distributed system maintenance with AWS Lambda and Kinesis, **lowering AWS expenses by 25%**
- Engineered **persistent storage architecture** linking Okta SSO authentication with AWS EFS, ECS, and RDS to support secure stateful development sessions across container restarts
- Designed and implemented Terraform-based infrastructure provisioning and Jenkins CI/CD pipelines, **decreasing deployment times by 66%**

Undergraduate Data Science Researcher | *Bayer Crop Sciences, Syngenta*

August 2023 – May 2025

The Data Mine, Purdue University | Python, R, Sci-kit learn, Pandas

West Lafayette, IN

- Trained a LightGBM model on **500k+ rows** to model climate change impact on corn yields at **91% accuracy**
- Optimized model performance through feature engineering and cross-validation, reducing RMSE by **24%**
- Implemented data preprocessing and QC pipelines for **100M+ data points**, validated using mixed linear models

PROJECTS

MyPort | *TypeScript, OpenAI, LangChain, Supabase* | [GitHub](#) | [Link](#)

October 2025

- Pioneered RAG portfolio chatbot with Supabase, OpenAI embeddings, and LangChain retrievers
- **Reduced latency by 20%** with vector indexing and chunked doc ingestion, **supporting idempotent updates**

Evallm | *TypeScript, React, Next.js, Vercel, OAuth2, MongoDB* | [GitHub](#) | [Link](#)

July 2025 – August 2025

- Architected and deployed full-stack platform for **benchmarking LLM outputs**, integrating model inference with experiment tracking and analytics pipelines
- Integrated **real-time model inference**, enabling low-latency feedback and interactive evaluation workflows

UNIX-like Shell | *C++, Lex/Yacc*

February 2025 – April 2025

- Implemented POSIX-style shell in C++ with advanced parsing (Lex/Yacc), supporting **I/O redirection, pipelines, subshell execution, and signal handling**
- Handled process management using fork/exec & pipe syscalls, enabling **concurrent processes** and zombie reaping
- Constructed command history and environment variable expansion with **O(1) lookup**

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, R

Cloud/Data: AWS, boto3, psycpg2, PostgreSQL, Pandas, Flyway

Dev Tools: Git, FastAPI, React, Node.js, Next.js, Vite, Tailwind, Docker, MongoDB, Terraform, Jenkins